

Scalability & Future Plan

Date	29 June 2026
Team ID	XXXXXX
Project Name	Rising Waters
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Static Dataset (No Real-Time Ingestion)	The model is trained on historical CSV data and cannot ingest live weather feeds, limiting its ability to respond to rapidly changing meteorological conditions.	High
2	Single-Model Inference Only	The system currently uses one saved XGBoost model with fixed hyperparameters. There is no automated retraining pipeline, so model accuracy may degrade as new weather patterns emerge.	High
3	Limited Input Feature Set	Predictions rely on only three meteorological features (annual rainfall, cloud visibility, seasonal rainfall). Additional factors like river water levels, soil saturation, and wind speed could improve accuracy.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single-user Flask development server; no load balancing or concurrent session handling.	Deploy on IBM Cloud Kubernetes service with horizontal pod auto-scaling to support hundreds of simultaneous users. Add a CDN and API gateway for traffic distribution.
2	Data Storage	Flat CSV files stored locally; no database integration.	Migrate to IBM Db2 or Cloudant NoSQL database for structured weather records. Integrate IBM Watson IoT Platform for real-time sensor data streaming and storage.

		Limited to historical batch data only.	
3	Performance	Inference is fast for single predictions but not optimised for batch or concurrent requests. No caching layer exists.	Add Redis caching for repeated queries. Containerise the model as a microservice using Docker. Enable batch prediction endpoints for processing multiple regions simultaneously.
4	Security	No authentication, authorisation, or API key management. Suitable only for internal demonstration use.	Implement OAuth 2.0 / IBM App ID for role-based access control. Add HTTPS enforcement, input validation, rate limiting, and audit logging for all prediction requests.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Real-Time Data Integration & Automated Retraining	Months 3–6	IoT and weather API feeds replace static CSV data. Model auto-retrains monthly on fresh data, maintaining prediction accuracy above 95% in production.
Phase 3	Expanded Feature Engineering & Multi-Model Ensemble	Months 7–12	Incorporates river levels, soil saturation, wind speed, and elevation data. An ensemble of XGBoost + LSTM (for time-series patterns) improves accuracy and handles seasonal drift.
Phase 4	Mobile App, Alert System & Multi-Language Support	Year 2	Native mobile application with push notifications for flood alerts. SMS gateway integration for low-connectivity regions. Multi-language UI to serve diverse communities across India.